MIL-C-48354 (PA)
15 January 1975

MILITARY SPECIFICATION
CARTRIDGE, IGNITION, XM702E2
LOADING, ASSEMBLING AND PACKING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the loading, assembling and packing of the XM702E2, Ignition Cartridge for 60mm mortar ammunition.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-223 - Powder, Black
MIL-P-46425 - Primer, Percussion, M35, Loading Assembling and Packing
MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for

STANDARDS

MILITARY

MIL-STD-286 - Propellants, Solid, Sampling, Examination and Testing.
MIL-STD-453 - Inspection, Radiographic

DRAWINGS

PICATINNY ARSENAL

FSC: 1310

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PRODUCT AND PACKAGE DRAWINGS

9285465 - Cartridge, Ignition, XM702E2
9287826 - Packing and Marking for Box, Fiberboard, for Cartridge Ignition, XM702E1 and XM702E2

INSPECTION EQUIPMENT DRAWINGS

9278622 - Envelope Drawing for Leak Testing of Ignition Cartridges

2.2 Other publications.-The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

Code of Federal Regulations

49 CFR 71-90 Interstate Commerce Commission Rules and Regulations for the Transportation of Explosive and Other Dangerous Articles.

(Interstate Commerce Commission Regulations are now a part of the Code of Federal Regulations (1949 Edition and revisions) available from the Superintendent of Documents, Government Printing Office, Washington 25, D.C. Orders for the above publications should cite "49 CFR 71-90 (latest revision)."

3. REQUIREMENTS

3.1 Materials.-Materials shall be in accordance with applicable drawings and specifications.

3.2 Assembly.-The ignition cartridge assembly shall comply with all requirements specified on Drawing (dwg) 9285465 and with all requirements specified in applicable specifications.

3.3 Moisture content of black powder pellet.-The moisture content of the black powder pellet (Dwg. 7549177) at the time of assembling into the ignition cartridge shall not exceed 0.50 percent. Testing shall be as specified in 4.4.3.1.

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3.4 Sensitivity

3.4.1 Non-functioning.-The primer shall not function under an applied energy value of 8.85 inch ounces minimum (min.) when tested as specified in 4.4.3.4.1.

3.4.2 Functioning.-The primer shall function under an applied energy value of 82.60 inch ounces maximum (max). when tested as specified in 4.4.3.4.2.

3.5 Leak test.-The loaded cartridges shall satisfactorily pass the leak test specified in 4.4.3.5.

3.6 Proving ground test.-The ignition cartridges shall comply with the following requirements (see 4.4.3.6).

3.6.1 Misfires.-The ignition cartridges shall function on the first attempt, without any audible hangfire.

3.6.2 Metal part security.-There shall be no evidence of rupture or metal parts separation in the gun bore or in flight.

3.6.3 Velocity and functioning (Proving ground).-The cartridge assemblies shall function and shall impart a corrected muzzle velocity of not more than 225 feet per second (fps) and not less than 205 fps. In addition, the velocity standard deviation of the lot shall be no greater than 2.4 fps.

3.7 Compatibility of tape at 90°C.-The reactivity value shall not exceed 5 ml of gas when determined as specified in 4.4.3.8.

3.8 Workmanship.-All parts and assemblies shall be fabricated and loaded in a thorough workmanlike manner. They shall be free of burrs, sharp edges, cracks, dirt, rust and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. All required markings shall be neat and sharply defined.

3.9 Radiographic examination.-The loaded ignition cartridges, when hand assembled, shall satisfactorily pass the X-Ray examination specified in 4.5.7. Ignition cartridges assembled on automated equipment need not be X-Rayed provided adequate and approved

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integral inspection means assure that assembly work performed conforms to the component drawing requirements and achieves the desired degree of completeness and freedom from evidence of poor workmanship.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions.-Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections.-The following types of inspection shall be conducted on this item:

a. First Article Inspection.
b. Quality Conformance Inspection.

4.3 First article inspection

4.3.1 Submission.-The contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with provisions of 4.3.2. Fifty (50) ignition cartridge assemblies shall be submitted for testing to a Government Proving Ground designated by the contracting officer. In addition, one hundred (100) assemblies shall be submitted for examination and tests at Picatinny Arsenal, Dover, New Jersey. All tests shall be performed with the equipment and in accordance with the methods, specified in 4.5.

4.3.2 Inspections to be performed.-The First Article parts submitted in accordance with paragraph 4.3.1 will be subjected by the Government to any or all the requirements of the applicable drawings and to the inspections and tests listed in paragraphs 4.4.2 and 4.4.3.

4.3.3 Rejection.-See MIL-A-48078 (PA).

4.4 Quality conformance inspection

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4.4.1 Inspection lot formation.-Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, inspection lots of ignition cartridges shall contain:

a. Metal parts of one lot interfix from one manufacturer.
b. Black powder from not more than one lot from one manufacturer.
c. Percussion elements from not more than one lot interfix from one manufacturer.
d. M9 propellant from not more than one lot from one manufacturer.
e. Tape from not more than one manufacturer or manufacturer part number.
f. Adhesive sealant from one manufacturer.
g. Paper parts of one lot interfix from one manufacturer.

4.4.2 Examination.-See MIL-A-48078.

a. Sampling plans. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS MIL-C-48354 (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9280540
4.4.2.1	Head, Primer				NEXT HIGHER ASSEMBLY 9285465
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major A 101.	Flash hole obsturcted		100%	3.2	Gage
Major B	None defined				
Minor	None defined				

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48354 (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 8880640
4.4.2.2	Pellet				NEXT HIGHER ASSEMBLY 9280529
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major A Ioi.	Weight of pellet		0.40%	3.2	Balance
Major B	None defined				
Minor	None defined				

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CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9280529
4.4.2.3	Flash Tube Pellet Loading Assembly				NEXT HIGHER ASSEMBLY 9285465
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical 1.	Flash holes not completely covered with tape		100%	3.2	Visual
Major A 101.	Pellet Missing		100%	3.2	Visual
Major B 131.	Pellet Not Flush or Below Flush		0.40%	3.2	Visual
Minor 201.	Tape covering overlap or location improper		0.65%	3.2	Visual
202.	Evidence of poor workmanship		0.65%	3.8	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48354 (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.4	Tube Assembly				9285301
					NEXT HIGHER ASSEMBLY 9285465
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major A	None defined				
Major B					
131.	Outside diameter, max		0.40%	3.2	Gage
132.	Inside diameter, min		0.40%	3.2	Gage
133.	Length of tape fold-in, max		0.40%	3.2	Gage
134.	Length of tape overlap, max		0.40%	3.2	Gage
135.	Complete coverage by tape without wrinkles or separation		0.40%	3.2	Visual
Minor					
201.	Evidence of poor workmanship		0.65%	3.8	Visual

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS MIL-C-48354 (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9285465
4.4.2.5	Cartridge, Ignition, Prior to Assembly of Firing Plug				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major A					
101.	Percussion element missing		100%	3.2	Visual
102.	Percussion element inverted		100%	3.2	Visual
103.	Gasket missing		100%	3.2	Visual
Major B					
131.	Lacquer seal missing or inadequate around percussion element		0.40%	3.2	Visual
132.	Percussion element above max		0.40%	3.2	Visual
Minor	None defined				

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CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.6	Cartridge, Ignition, Prior to Assembly of Flash Tube Pellet Loading Assembly	1 OF 1	9285465
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major A 101.	Pellet cavity coating missing, inadequate or incompletely dry prior to assembling flash tube assembly		0.40%	3.2	Visual
Major B	None defined				
Minor	None defined				

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48354 (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.7	Cartridge, Ignition, Prior to Assembly of Cap				9285465
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1.	Flash tube assembly missing		100%	3.2	Visual
Major A					
101.	Liner missing		100%	3.2	Visual
102.	Propellant missing		100%	3.2	Visual
Major B	None defined				
Minor					
201.	Foreign matter in charge		0.65%	3.2	Visual

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CLASSIFICATION OF DEFECTS & TESTS MIL-C-48354 (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.8	Cartridge, Ignition				9285465
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1.	Firing plug not loose after assembly		100%	3.2	Manual
Major A	None defined				
Major B					
131.	RTV sealant missing, inadequate or excessive		0.40%	3.2	Visual
132.	Length from shoulder to end, max		0.40%	3.2	Gage
133.	Support fails to withstand reapplication of minimum assemble torque before pinning		0.40%	3.2	Test
134.	Support pins missing		0.40%	3.2	Visual
Minor					
201.	Diameter		0.65%	3.2	Gage
202.	True position		0.65%	3.2	Gage
203.	Marking missing, misleading or unidentifiable		0.65%	3.2	Visual
204.	Evidence of poor workmanship		0.65%	3.8	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48354 (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.9	Packed Carton, Prior to Sealing				9287826
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major A	None defined				
Major B					
131.	Tray broken, split or incomplete		0.40%	5.1	Visual
132.	Improperly packed		0.40%	5.2	Visual
Minor					
201.	Cartridges missing		0.65%	5.2	Visual

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CLASSIFICATION OF DEFECTS & TESTS MIL-C-48354 (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.10	Carton After Sealing				9287826 NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major A	None defined				
Major B					
131.	Cartons damaged (cut or damaged through all layers)		0.40%	5.2	Visual
132.	Carton not properly sealed		0.40%	5.2	Visual
Minor					
201.	Contents loose		0.65%	5.2	Visual
202.	Marking missing, incorrect or illegible		0.65%	5.2	Visual

NOTES:

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4.4.3 Testing

4.4.3.1 Moisture content of black powder(see 3.3) Major A defect.-A sample of twenty-five black powder pellets shall be randomly selected from each eight hours production and shall be tested as specified in 4.5.1. Any failure to comply with the applicable requirements shall be cause for rejection of the eight hours production represented by the sample. If loading has begun, ignition cartridges loaded with the non-conforming black powder shall be rejected.

4.4.3.2 Weight of propellant (see dwg. 9285465) weight above max, Major A defect. Weight below minimum (min) Critical defect.-The propellant charge shall be weighed or volumetrically dispensed into a tared container and then check-weighed 100 percent. The initial weighing and the check-weighing shall be performed by the different operators on different scales. Weight limits for check weighing should be established within the applicable weight requirement, depending on the accuracy and precision of the weighing system, to assure compliance with charge weight requirements. Any charge failing to meet the applicable requirement shall be considered defective and removed from the lot.

4.4.3.3 Weight of propellant in cartridge assembly (see dwg. 9285465) weight above max, Major A defect. Weight below minimum (min), Critical defect.

4.4.3.3.1 315 sample cartridge assemblies shall be selected from each lot for this test. If four or more samples exceed the maximum charge weight or if any sample is below the minimum charge weight, the lot shall be rejected. Testing shall be as specified in 4.5.4.

4.4.3.3.2 Alternate procedure.-Loaded ignition cartridges shall be weighed 100 percent and the net weight of propellant determined by subtracting parts tare weight from the gross weight. The new weight of propellant should compare with the original weight of propellant within $\pm$ 0.5 grain.

4.4.3.4 Sensitivity

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4.4.3.4.1 Non-functioning (see 3.4.1) Critical defect.- Beginning with the first lot produced and continying until three consecutive lots have complied with the acceptance criteria specified, 200 primed heads, assembled with firing plugs, shall be randomly selected from each lot for this test. If one or more primed heads fail to comply with the specified requirement, the lot shall be rejected. The test shall be performed as specified in 4.5.2.1.

4.4.3.4.1.1 After three consecutive lots have met the criteria of 4.4.3.4.1, 125 primed heads, assembled with firing plugs, shall be randomly selected from each lot for this test. If one or more primed heads fail to comply with the specified requirement, the lot shall be rejected. The test shall be performed as specified in 4.5.2.1.

4.4.3.4.2 Functioning (see 3.4.2) Major A defect.-Beginning with the first lot produced and continuing until three consecutive lots have complied with the acceptance criteria specified, 500 primed heads, assembled with firing plugs, shall be randomly selected from each lot for this test. If 4 or more primed heads fail to comply with the specified requirement, the lot shall be rejected. If 2 or 3 defectives are found a second sample of 500 primed heads, shall be tested. If the combined number of defectives in both the first and second sample is 5 or more, the lot shall be rejected. Any primed heads, in the randomly selected sample missing a component or with an incorrectly assembled component which will result in an end item misfire shall be considered defective and shall be included in the rejection criteria. The test shall be performed as specified in 4.5.2.2.

4.4.3.4.2.1 After three consecutive lots have met the criteria of 4.4.3.4.2, a sample of 315 primed heads, assembled with firing plugs, shall be randomly selected from each lot for this test. If 3 or more primed heads fail to comply with the specified requirement, the lot shall be rejected. If 1 or 2 defectives are found a second sample of 315 primed heads, shall be tested. If the combined number of defectives in both the first and second sample is 4 or more the lot shall be rejected. Any

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primed heads, in the randomly selected sample missing a component or with an incorrectly assembled component which will result in an end item misfire shall be considered defective and shall be included in the rejection criteria. The test shall be performed as specified in 4.5.2.2.

4.4.3.5 Leak test (see 3.5) Major A defect.-The ignition cartridges shall be tested 100 percent. Any ignition cartridge which fails to comply with the applicable requirement shall be classed defective and removed from the lot. The test shall be performed as specified in 4.5.3.

4.4.3.6 Velocity, functioning and security (see 3.6) Major A defect.

4.4.3.6.1 Beginning with the first lot produced and continuing until three consecutive lots have complied with the applicable requirements specified, the sampling plan and procedure for this test shall be in accordance with MIL-STD-414, Section B, Table B-3, Code Letter M and an AQL of 1.0 percent. In addition, if the velocity standard deviation of the sample multiplied by the factor 0.81 exceeds the lot requirement, the lot shall be rejected. If any round fails to function for reasons attributable to ignition cartridge failure or if a metal parts separation occurs, the lot shall be rejected. The test shall be performed as specified in 4.5.5.

4.4.3.6.2 After three consecutive lots have met the criteria of 4.4.3.6.1, the sampling plan for this test shall be in accordance with MIL-STD-414, Section B, Table B-3, Code Letter I, and an AQL of 1.0 percent. In addition, if the velocity standard deviation of the sample multiplied by the factor 0.75 exceeds the applicable lot requirement, the lot shall be rejected. If any round fails to function for reasons attributable to ignition cartridge failure or if a metal parts separation occurs, the lot shall be rejected. The test shall be performed as specified in 4.5.5.

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4.4.3.7 Check test for deterioration of Primer, Percussion M35.-If the total elapsed time between orignial acceptance of the M35 primer lot and the assembly of that lot into the ignition cartridge exceeds two years, or if the M35 primers have been subjected to adverse conditions for any period of time since previous test, the M35 primer lot shall be subjected to and must satisfactorily pass the check test for deterioration specified in Specification MIL-P-46425 immediately before the M35 primer lot is assembled into the ignition cartridges.

4.4.3.8 Compatibility of tape at 90°C (see 3.7) Critical defect.-A sample from every lot of tape supplied by one manufacturer shall be randomly selected and tested as specified in 4.5.6. If any sample fails to comply with the applicable requirement, the lot shall be rejected.

4.4.3.9 Radiographic examination (see 3.9) Critical defect.-The ignition cartridges shall be X-Rayed 100 percent. Any cartridges which fail to comply with the applicable requirement shall be classed defective and removed from the lot. The test shall be performed as specified in 4.5.7.

4.4.4 Inspection equipment.-The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with 4.4.2.1 and 4.5 (see 6.2).

4.5 Test methods and procedures

4.5.1 Moisture content of black powder.-The moisture content of black powder shall be determined in accordance with the procedure specified in Specification MIL-P-223.

4.5.2 Sensitivity

4.5.2.1 Non-functioning.-The primed head, assembled with firing plug, shall be assembled to a suitable fixture. The firing pin of the test fixture shall be adjusted so that impact will be on the center of the firing plug of the primer. A weight of 2.95

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pounds, minimum, shall be dropped on to the firing pin from a height of 3/16 inch, minimum, as if to function the primer. The height of drop shall be measured as the height from the top of the firing pin to the bottom of the weight. Any primer which functions shall be classed defective. Metal parts used in this test shall not be returned to the lot.

4.5.2.2 Functioning.-The primed head, assembled with firing plug, shall be assembled to an appropriate test fixture. The firing pin of the test fixture shall be adjusted so that impact will be 0.05 inch off center of the firing plug of the primer. A weight of 2.95 pound, maximum, shall be dropped on the firing pin to function the primer from a height of 1 3/4 inches, maximum. The height of drop shall be measured as the height from the top of the firing pin to the bottom of the weight. Any primer which fails to function shall be classed defective. Metal parts used in this test shall not be returned to the lot.

4.5.3 Leak test.-The ignition cartridges shall be tested in accordance with drawing 9278622.

4.5.4 Weight of propellant.-The sample cartridge assemblies shall be cut open using a method that will not generate chips and all propellant removed and placed on the pan of a precision or analytical balance. The weight of propellant from each sample shall be determined to the nearest 0.01 grain.

4.5.5 Proving ground test.-The test shall be performed at a Government Proving Ground in accordance with the applicable USATECOM Acceptance Test Procedure.

The sample ignition cartridges shall be assembled with standard finned projectiles (3.90 pounds (lbs)) for which they were manufactured. The supports of the test samples (only) shall be marked with an ink marker for positive identification in the event of a separation. The complete assembly, less propelling charges, shall be fired from a standard 60mm, M19, mortar at an elevation of forty-five degrees.

Reference rounds shall be fired without propelling charges for the purpose of velocity correction. At least one reference round should be fired for every three test rounds.

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Individual velocities shall be recorded on the firing record for both reference and test rounds. Velocity standard deviations and arithmetic averages shall be calculated and recorded. Temperature of the day and time of firing shall also be recorded. In the event of a misfire, the misfired ignition cartridge shall not be disassembled by the proving ground but shall be forwarded to the responsible technical activity.

4.5.5.1 Refiring.-If for any reason the proving ground considers that test conditions have detrimentally affected the test results, additional cartridges as required shall be tested.

4.5.6 Determination of compatibility of tape at 90°C.- The reactivity value shall be determined in accordance with Method 403.1.2 of MIL-STD-286 with the following exceptions:

Paragraph 2.1-The specimens shall consist of 5.00 grams of propellant and 5.00 grams of tape.

Paragraph 4.1.2-After determining the volume of gas evolved from each heating tube, the reactivity value is calculated as follows:

$R = C - (A + B)$

A = amount of gas evolved by the propellant
B = amount of gas evolved by the tape
C = amount of gas evolved by the 50/50 mixture or propellant and tape.
R = reactivity value.

Paragraph 4.3-Place a weighed portion of 2.5 grams of the propellant in one heating tube, and 2.5 grams of tape in another heating tube. In a third heating tube place a 5 gram portion of a 50/50 mixture of the tape and propellant.

4.5.7 Radiographic examination.-The ignition cartridges shall be examined in accordance with MIL-STD-453 procedure. The items shall be X-rayed with the radiographic beam perpendicular to the longitudinal axis of the ignition cartridge. The radiograph shall be examined to determine presence of all parts and propellant, and proper orientation of firing plug, primer, and flash tube assembly. The penetrameter

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used in the application specified in MIL-STD-453 shall be designed and constructed to insure a 2 percent sensitivity or better. In the event an original X-ray picture leaves doubt as to acceptability, the ignition cartridge shall be re-X-rayed at the angle 90 degrees from that of the original exposure and in the same plane. If any ignition cartridge shows evidence of missing or misoriented parts, it shall be classed defective and removed from the lot.

4.5.7.1 Radiographic evaluation.-The ignition cartridges submitted as the First Article Samples, shall be radiographed in accordance with 4.5.7 and the radiographs shall be submitted to the Government for evaluation.

5. PREPARATION FOR DELIVERY

5.1 Packaging.-Packaging shall be as specified on Drawing 9287826.

5.2 Packing and marking.-Packing and marking shall be as specified on Drawing 9287826.

6. NOTES

6.1 Ordering data.-See MIL-A-48078 (PA).

a. Ammunition data cards shall be prepared in accordance with MIL-STD-1167.

6.2 Submission of designs for approval.-See MIL-A-48078. Submit equipment designs, as required, to Commander, Picatinny Arsenal, ATTN: SARPA-QA-T-T, Dover, New Jersey 07801.

6.3 New material test sample requirements.-All requests for permission to use materials other than those specified on applicable drawings or specifications should be accompanied by a sample of the proposed substitute material to permit evaluation by Picatinny Arsenal. Prior to approval, all substitute materials must be tested for compatibility with M9 Propellant in addition to such testing required to evaluate the performance of the proposed substitute.

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6.4 Submission of test data.-In addition to the normal distribution of records, when the ignition cartridge is procured by the Department of the Army, one copy of all ballistic test data shall be forwarded to the following:

Commander
Picatinny Arsenal
ATTN: SARPA-QA-A-A
Dover, New Jersey 07801

6.5 Check test for possible deterioration of Cartridge, Ignition.-The following test is designed to prevent the loading of major ammunition items with cartridges which have deteriorated. The test shall be performed on each lot of cartridges which has been stored under normal conditions for more than two years or under adverse conditions for any period of time. If the specification for the ammunition item into which the cartridge is assembled does not require that the check test be performed, the contracting officer shall insert a provision for the performance of the check test in the contract. The contracting officer shall arrange to have the contractor reimbursed for any expense incurred in the performance of the check test for possible deterioration.

6.5.1 Velocity, functioning and security.-A sample from each lot shall be selected in accordance with Standard MIL-STD-414, Section B, Table B-3, Code Letter M and an AQL of 1.0 percent. In addition, if the velocity standard deviation of the sample multiplied by the factor 0.81 exceeds the lot requirement, the lot shall be rejected. If any round fails to function for reasons attributable to ignition cartridge failure, or if a metal parts separation occurs, the lot shall be rejected. The test shall be performed as specified in 4.5.5.

Custodian:
ARMY-PA

Preparing Activity:
ARMY-PA

Project Number: 1310-A057

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